

Pediatric DKA



Section I: Scenario Demographics

Scenario Title:	DKA and Decreased LOC		
Date of Development:	09/05/2016 (DD/MM/YYYY)		
Target Learning Group:	☐ Juniors (PGY 1 - 2)	☒ Seniors (PGY $\geq$ 3)	☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Donika Orlich (adapted from case by Lindsey McMurray)
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To prioritize the management of an acutely unwell child.
CRM Objectives:	1) Employs good communication skills by taking verbal handover from the sending facility and delivering clear instructions prior to patient transfer 2) Manages an anxious parent witnessing a resuscitation
Medical Objectives:	1) Implements an EBM approach to resuscitation and treatment of pediatric DKA 2) Recognizes the need for IO access after failure to obtain peripheral IV 3) Recognizes cerebral edema and implements timely treatment 4) Anticipates decline in a DKA patient requiring intubation and plans accordingly

Case Summary: Brief Summary of Case Progression and Major Events

The learners receive a call from a peripheral hospital about transferring an unwell 8-year-old girl with new DKA. She has been incorrectly managed, receiving a 20cc/kg bolus for initial hypotension as well as an insulin bolus of 8 units (adult sliding scale dose for glucose of >20). The learner must perform a telephone consultation and dictate new orders. On arrival, EMS will state that they lost the IV en route, and the patient will become more somnolent in the ED. The learner should begin empiric treatment for likely cerebral edema and concurrently manage the DKA. Physical exam will show a peritonitic abdomen with guarding in the RLQ. Empiric Abx should be started for likely appendicitis. Due to decreasing neurologic status and vomiting, the patient will eventually require an advanced airway. The challenge is to optimize the peri-intubation course and ventilation to allow for compensation of her metabolic acidosis.

References

- Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby. Chapter 126: Diabetes Mellitus and Disorders of Glucose Homeostasis.
- EMCrit (2009). Intubating the patient with severe metabolic acidosis. Accessed on September 11, 2015 at <http://emcrit.org/podcasts/tube-severe-acidosis/>



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Section IV: Scenario Script

A. Clinical Vignette: To Read Aloud at Beginning of Case			
<i>Outside Patch: We have an 8-year-old female we want to send for DKA. She presented after feeling generally "unwell" for 3 days, with some accompanying abdominal pain and vomiting. Her blood glucose came back at 24 with a pH of 7.15 and HCO₃ of 12, so we made the diagnosis of DKA. She received a 20mL/kg bolus for hypotension (BP 90/60) and Humulin R 8 unit bolus (as per our hospital sliding scale). What do you want for insulin and fluids before we send her?</i> <i>If asked:</i> <i>K = 4.5. No other PMHx. Also did AXR which was normal (very tender abdomen).</i>			
B. Scenario Cast & Realism			
Patient:	☒ Pediatric Computerized Mannequin ☐ Mannequin ☐ Standardized Patient ☐ Hybrid ☐ Task Trainer	Realism: <i>Select most important dimension(s)</i>	☒ Conceptual ☐ Physical ☐ Emotional/Experiential ☐ Other: ☐ N/A
Confederates	Brief Description of Role		
Mother	Gives past medical history and history of presenting complaint. Becomes agitated if not addressed.		
EMS	Handover care on arrival. Lost IV en route.		
C. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
D. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☒ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other:	
☒ Intraosseous Set-up	☐ LMA	☐ Other:	
E. Moulage			
None required.			
F. Approximate Timing			
Set-Up:	5 min	Scenario:	15 min
		Debriefing:	30 min

Section V: Patient Data and Baseline State



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A. Patient Profile and History					
Patient Name: Anna		Age: 8		Weight: 25kg	
Gender: ☐ M ☒ F		Code Status: Full			
Chief Complaint: Altered LOC					
History of Presenting Illness: Tired and "unwell" for 3 days with some mild abdominal pain and vomiting. Febrile today so her parents brought her to their local ED.					
Past Medical History:	nil		Medications:	nil	
	Immunizations UTD				
Allergies: nil					
Social History: lives with mom and dad and 1 older brother.					
Family History: no family history.					
Review of Systems:	CNS:	Tired. EMS state confused en route, asking for her dog repeatedly.			
	HEENT:	No HA. No neck stiffness.			
	CVS:	No complaints			
	RESP:	No cough, sputum. No SOB			
	GI:	Abdominal pain. Crying during ambulance ride due to pain.			
	GU:	Polyuria.			
MSK:	Normal.		INT:	Normal.	
B. Baseline Simulator State and Physical Exam					
☒ No Monitor Display		☐ Monitor On, no data displayed		☐ Monitor on Standard Display	
HR: 120/min	BP: 95/60	RR: 34/min (deep)	O ₂ SAT: 99%		
Rhythm: NSR	T: 38.5°C	Glucose: HIGH	GCS: 14 (E4 V4 M6)		
General Status: Appears unwell.					
CNS:	Eyes open to pain, obeys commands.				
HEENT:	Acetone breath, dry mucous membranes. No nuchal rigidity				
CVS:	Tachycardic. Brisk capillary refill.				
RESP:	High RR with deep Kusmaul breathing.				
ABDO:	++ RLQ tenderness with rebound and guarding.				
GU:	Normal.				
MSK:	Normal.		SKIN:	Normal.	

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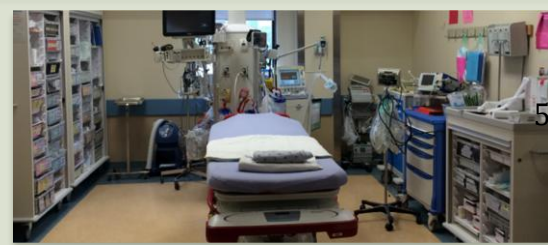


Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
Criti-call Patch Rhythm: Sinus tach HR: 140 → 120 (after 20cc/kg bolus) BP: 90/60 → 95/60 (after bolus) RR: 34/min O ₂ SAT: 99% T: 38.5°C	GCS 14 (confused) Given 2mg morphine for abdominal pain (now settled)	<u>Learner Actions</u> - ☐ Ask for information around labs and treatments initiated - ☐ Patient PMx, meds, allergies - ☐ Ask if she is voiding (she is) - ☐ Insulin infusion at 0.05-0.1u/kg/hr - ☐ IV NS + 40mmol KCl/L at 3-5mL/kg/hr - ☐ Provide feedback to sending MD re: insulin dosing & IV fluid in pediatric DKA	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - No other probing by learner → MD to ask “would you like to know anything else?” <u>Triggers</u> <i>For progression to next state</i> - 2 minutes → “assume 30 minutes have passed” → 1. EMS Arrival
1. EMS Arrival Rhythm: Sinus tach HR: 120/min BP: 95/60 RR: 34/min O ₂ SAT: 99% T: 38.5°C	States begins with EMS handover (lost IV en route) GCS 14 (confused) Cries when RLQ palpated.	<u>Learner Actions</u> - ☐ Monitor, full vitals - ☐ Check glucose (HIGH) - ☐ 2 large bore IVs (unable) - ☐ IO access with local anesthetic - ☐ Take history from & reassure mom - ☐ Blood work (standard + VBG, lactate, serum ketones, osmols, lytes, blood cultures, urine R&M) - ☐ IV NS + 40mmol KCl/L at 3-5mL/kg/hr - ☐ IV insulin at 0.05-0.1 U/kg/hr - ☐ Administer antibiotics (eg - ancef + flagyl)	<u>Modifiers</u> - IV attempt → unsuccessful x2 - Nurse runs VBG immediately after ordered and gives results to team leader - If mom not addressed → mom to become more intrusive and difficult to have in room <u>Triggers</u> - 5 minutes → 2. Deterioration
2. Deterioration Rhythm: Sinus tach HR: 110/min BP: 90/50 RR: 28/min O ₂ SAT: 98%	GCS 13- E3V4M6 Patient is more confused, complains of headache.	<u>Learner Actions</u> - ☐ Re-check glucose (24) - ☐ Decrease insulin infusion by half (0.025-0.05 U/kg/hr) - ☐ Decrease IV fluid maintenance by half - ☐ Mannitol at 1-2mL/kg OR hypertonic saline 3% at 3ml/kg for possible cerebral edema - ☐ Elevate HOB	<u>Modifiers</u> - Mother not updated → becomes obstructive around daughter and demands 2 nd opinion <u>Triggers</u> - 2 minutes into state or all actions complete → 3. Obtunded



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3. Obtunded Rhythm: NSR HR: 70/min BP: 140/95 RR: 28/min O ₂ SAT: 97% T: 36.5°C	GCS 6-E1V1M4 Vomiting and gagging on secretions.	<u>Learner Actions</u> - ☐ Recovery position, suction - ☐ Plan for intubation - ☐ ± Pre-intubation bicarbonate - ☐ ± Pre-intubation 10cc/kg fluid bolus - ☐ Push-dose vasopressor at bedside - ☐ Set vent to match patient RR (and VBG to match pCO ₂)	<u>Modifiers</u> - No intubation by 2 min into state → O ₂ SAT slowly to 88% <u>Triggers</u> - Intubation with paralytic & resp rate not considered → 4. PEA Arrest - Intubation with resp rate considered → 5. Resolution
4. PEA Arrest Rhythm: PEA HR: no pulse BP: not detectable RR: BVM rate O ₂ SAT: no tracing T: 36.5°C	Non-responsive and pulseless.	<u>Learner Actions</u> - ☐ Ensure quality CPR (15:2) - ☐ Epinephrine at 0.01mg/kg (.25mg) q 3-5 minutes. - ☐ Sodium bicarbonate (1meq/kg/dose ~ ½ amp) - ☐ Confirm tube placement - ☐ Go through Hs, Ts	<u>Modifiers</u> - If needed, after 1 cycle, RT will suggest ventilating at a faster rate to match pre-intubation RR <u>Triggers</u> - One CPR cycle after RR increased → 5. Resolution
5. Resolution Rhythm: Sinus tach HR: 140/min BP: 90/50 RR: 34/min (tubed) O ₂ SAT: 99% T: 36.5°C	GCS 3T	<u>Learner Actions</u> - ☐ Call PICU and Gen Sx - ☐ Re-check electrolytes, gas - ☐ Post-intubation CXR - ☐ Insert NG/OG - ☐ Initiate sedation - ☐ ± Cooling (if had arrest)	<u>END CASE WITH PICU ARRIVAL</u>



Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results						
Na: 128	K: 4.5	Cl: 100	HCO ₃ : 8	BUN: 34	Cr: 60	Glu: 31
Ca: n/a		Mg: n/a		PO ₄ : n/a		Albumin: n/a
VBG	pH: 6.9	PCO ₂ : 20	PO ₂ : 50	HCO ₃ : 8	Lactate: 5	
WBC: 16		Hg: 127		Hct: 0.400		Plt: 400

Images (ECGs, CXRs, etc.)
CXR post intubation
Source: https://emcow.files.wordpress.com/2012/11/normal-intubation2.jpg



Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
Educational Goal:	To prioritize the management of an acutely unwell child
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Sample Questions for Debriefing	
1) How do you feel that your team communicated throughout this case? 2) How did it feel to have the mother in the room? Do you think the team handled this well? What are some approaches for having parents at the bedside during a resuscitation? 3) What is the differential for a child with altered LOC? 4) What is your initial management of DKA in a child and how does it differ from an adult? 5) What special considerations must you keep in mind when intubating a DKA patient or any patient with severe acidosis of any etiology? 6) If you suspect a PEA arrest secondary to acidosis, what are your treatment options?	
Key Moments	
Recognition of initial incorrect management of pediatric DKA and acting as consultant to implement correct insulin and IVF therapy prior to transfer.	
Recognition of likely cerebral edema and implementing treatment (mannitol) and change to insulin/IVF therapy	
Recognizing the need to intubate, and vocalizing special considerations given severe acidosis.	
Running a PEA arrest (if applicable)	